

Simultaneous equations and inequalities - Set 2

TMUA practice questions by topic. Specification MM1.4, MM1.5, M4.16, M4.17. 7 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Which of the following inequalities has exactly the same solution set as $\frac{1}{x} > 2$?
- A $x > \frac{1}{2}$
 - B $0 < x < \frac{1}{2}$
 - C $x < \frac{1}{2}$
 - D $x < 0$ or $x > \frac{1}{2}$
 - E $x > 2$
 - F $0 < x < 2$
- 2 Find the complete set of values of x for which $(x - 1)(x - 3)(x - 5) > 0$.
- A $1 < x < 3$ or $x > 5$
 - B $x > 5$
 - C $1 < x < 5$
 - D $x < 1$ or $3 < x < 5$
 - E $x > 1$
 - F $x < 1$ or $x > 5$
- 3 Given that $|2x - 3| < 5$, which of the following statements is/are true?
- I $-1 < x < 4$
 - II $x^2 < 16$
 - III $x > -1$
- A none of them
 - B I only
 - C II only
 - D III only
 - E I and II only
 - F I and III only
 - G II and III only
 - H I, II and III

- 4 How many integers x satisfy $x^2 < 5x + 24$ and $x^2 > 3x$?
- A 4
B 8
C 5
D 7
E 6
F 9
- 5 The solution set of $ax^2 + bx + c < 0$ (with $a > 0$) is $-2 < x < 5$. What is the solution set of $ax^2 - bx + c < 0$?
- A $-2 < x < 5$
B $x < -5$ or $x > 2$
C $2 < x < 5$
D $x < -2$ or $x > 5$
E $-5 < x < -2$
F $-5 < x < 2$
- 6 Find the complete set of values of x for which $\frac{x-2}{x+1} \leq 0$.
- A $-1 \leq x \leq 2$
B $x < -1$ or $x \geq 2$
C $-1 < x \leq 2$
D $x \leq 2$
E $x \leq -1$ or $x \geq 2$
F $-2 \leq x < 1$
- 7 Consider the statement: for all real x , if $x > 2$ then $x^2 - 3x + 2 > 0$. Which of the following is true?
- A The statement is true, and so is its converse.
B The statement is true, but its converse is false.
C The statement is false, but its converse is true.
D The statement and its converse are both false.